

1.5 Classwork: Box-and-Whisker Plots — Markscheme

Calculator allowed. Marks are not printed on the student sheet; they are recorded here. Total: 19 marks.

Throughout: a whisker runs to the most extreme value that is not an outlier, and each outlier is drawn as a separate cross. The most likely error is a whisker drawn all the way to the outlier (or to the fence value $Q_3 + 1.5 \times IQR$, which is not a data value).

1. [Maximum mark: 9]

Ordered: 12, 15, 18, 20, 21, 22, 24, **25**, 26, 28, 30, 31, 33, 35, 58 ($n = 15$).

GDC: enter the fifteen times as a list and run **One-Variable Statistics**. The TI-Nspire excludes the median from each half, so Q_1 is the median of the lowest seven values and Q_3 the median of the highest seven.

(a) Median = 25 A1

$Q_1 = 20$, $Q_3 = 31$ A1

Both quartiles are needed for the second A1. Accept $Q_1 = 20.5$ and $Q_3 = 30.5$ from a hand method that includes the median in each half; feedback: the GDC convention is the one to use.

(b) $IQR = 31 - 20 = 11$ A1

FT: award A1 for $Q_3 - Q_1$ from the candidate's own quartiles. The range, $58 - 12 = 46$, earns no mark.

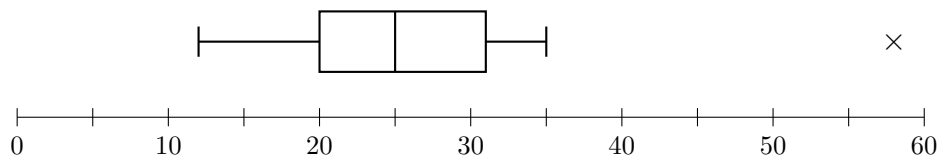
(c) $Q_3 + 1.5 \times IQR = 31 + 1.5 \times 11$ M1

$= 47.5$ A1

$58 > 47.5$, so 58 is an outlier. R1

M1 for $1.5 \times$ their IQR added to their Q_3 . R1 for the comparison with 58 stated, not just the fence value. Adding $1.5 \times IQR$ to the median or the mean earns no marks. The lower fence, $20 - 16.5 = 3.5$, is not required. FT: award all three marks from the candidate's own Q_3 and IQR, provided their fence is below 58.

(d)



Box from $Q_1 = 20$ to $Q_3 = 31$ with the median line at 25 A1

Whiskers to 12 and 35 A1

Outlier 58 marked with a cross A1

Award each mark independently; plotted values within half a small division (± 1 minute) are accepted. A whisker drawn to 58 or to 47.5 loses the second A1. FT: award the first A1 for a box drawn at the candidate's own quartiles from (a).

2. [Maximum mark: 10]

Class A (from the plot): minimum 40, $Q_1 = 52$, median 60, $Q_3 = 68$, maximum 84.

(a) Median = 60 A1

IQR = $68 - 52 = 16$ A1

The marks are independent. The range, $84 - 40 = 44$, earns no mark for the IQR.

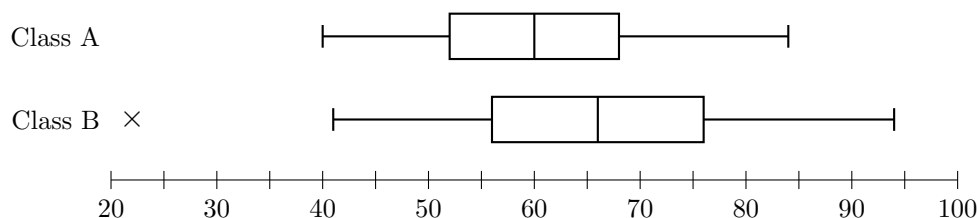
(b) Class B: IQR = $76 - 56 = 20$, $1.5 \times 20 = 30$ M1

Lower fence $56 - 30 = 26$, upper fence $76 + 30 = 106$ A1

$22 < 26$, so 22 is an outlier; $94 < 106$, so 94 is not. R1

M1 for $1.5 \times$ the Class B IQR. A1 for both fences. R1 for both conclusions, each linked to its fence. Using the Class A IQR earns no marks.

(c)



Box from 56 to 76 with the median line at 66 A1

Whiskers to 41 and 94 A1

Outlier 22 marked with a cross A1

Award each mark independently; values within ± 1 are accepted. A lower whisker drawn to 22 or to the fence 26 loses the second A1. FT: a candidate who concluded in (b) that 22 is not an outlier and draws the whisker to 22 with no cross earns the second A1 but not the third.

(d) Centre: the Class B median (66) is higher than the Class A median (60), so Class B generally scored higher. A1

Spread: the Class B IQR (20) is larger than the Class A IQR (16), so the Class B scores are more spread out. A1

Each A1 needs a direction stated and the values (or the statistic) quoted. Accept the ranges, 72 against 44, for the spread mark; feedback: the IQR is the better measure here because the range is driven by the Class B outlier. A comparison of means earns no mark — the means are not known. FT: award the spread mark from the candidate's own Class A IQR in (a).